Incidencia normal:

a) Obtener impedancias y constantes de propagación en ambos medios

b,c y d) Obtener expresiones fasoriales de todos los campos (i, r, t)

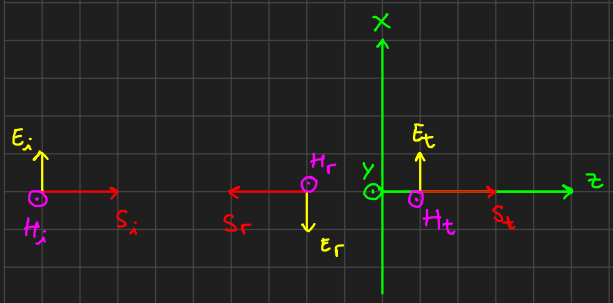
e) Obtener densidad de potencia promedio temporal (i, r, t) en Z=0. Verificar Balance de potencia

f) Obtener expresión de la densidad de potencia promedio temporal en función de z.

¿Para qué z $\langle P_t(z)\rangle / \langle P_t(z=0)\rangle$ < -100 dB? → transmitida?

medio 1 : aire $\varepsilon_1 = \varepsilon_0$, $\sigma_1 = 0$, $\mu_1 = \mu_0$

medio 2 : carbón $\varepsilon_2 = 10\,\varepsilon_0$, $\sigma_2 = 1k\ \frac{S}{m}$, $\mu_2 = \mu_0$



a) Medio ①

$$z_1 = \sqrt{\frac{j\omega\mu_1}{\sigma_1 + j\omega\varepsilon_1}} \overset{\sigma_1 = 0}{=} \sqrt{\frac{\mu_1}{\varepsilon_1}} = \sqrt{\frac{\mu_0}{\varepsilon_0}} = z_{00} = 120\pi\,\Omega \approx \boxed{377\,\Omega}$$

$$\gamma = \alpha + j\beta$$

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right)} \overset{\sigma_1 = 0}{=} \omega \cdot 0 = 0\ \frac{1}{m}$$

$$\beta = \omega \sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1\right)} \overset{\sigma_1 = 0}{=} \omega\sqrt{\mu_1\varepsilon_1} = 2\pi f \sqrt{\mu_0\varepsilon_0}$$

$$\beta = 2\pi \cdot 100\,MHz \sqrt{4\pi \cdot 10^{-7}\,\frac{H}{m} \cdot 8{,}85 \cdot 10^{-12}\,\frac{F}{m}} \approx 2{,}096\ \frac{1}{m}$$

$$\Rightarrow \boxed{\gamma_1 = j\,2{,}096\ \frac{1}{m}}$$

medio ②

$$z_2 = \sqrt{\frac{j\omega\mu_2}{\sigma_2 + j\omega\varepsilon_2}} =$$

aproximo: $\sigma_2 >> \omega\varepsilon_2$

$$\Rightarrow z_2 \approx \sqrt{j}\cdot\sqrt{\frac{\omega\mu_2}{\sigma_2}} = (1+j)\sqrt{\frac{\omega\mu_2}{2\sigma_2}} \approx \boxed{628{,}3\,(1+j)\ m\Omega}$$

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right)} \approx 628{,}3\ \frac{1}{m}$$

$$\beta \approx 628{,}3\ \frac{1}{m}$$

$$\boxed{\gamma \approx 628{,}3\,(1+j)\ \frac{1}{m}}$$

b) $\vec{E}_i$, $\vec{H}_i$, $\vec{S}_i$ fasoriales.

En el tiempo:

$$\vec{E}_i(z,t) = Re\left\{ E_{i0}\,\check{e}_i\, e^{j(\omega t - \vec{k}\cdot\vec{r})}\, e^{-\alpha z} \right\} \qquad (\alpha = 0)$$

$$\vec{k} = \beta\check{z} \approx 2\check{z}$$

$$\vec{E}_i(z,t) = Re\left\{ 1\ \frac{V}{m}\ \check{x}\ e^{j\left(628\cdot 10^6\ \frac{1}{s}\,t - 2z\right)} \right\}$$

⟶ fasorial: $\boxed{\vec{E}_i(z) = 1\ \frac{V}{m}\ \check{x}\ e^{-j2z}}$

$$\vec{H}_i(z,t) = Re\left\{ \frac{E_{i0}}{z_1}\,\check{h}_i\, e^{j(\omega t - \vec{k}\cdot\vec{r})}\, e^{-\alpha z} \right\}$$

$$\vec{H}_i(z,t) = Re\left\{ 2{,}65\ \frac{mA}{m}\ \check{y}\ e^{j\left(628\cdot 10^6\ \frac{1}{s}\,t - 2z\right)} \right\}$$

⟶ fasorial: $\boxed{\vec{H}_i(z) = 2{,}65\ \frac{mA}{m}\ \check{y}\ e^{-j2z}}$

$$\vec{S}_i = \vec{E}_i \times \vec{H}_i$$

$$\vec{S}_i = E_{i0}\,\check{x}\; e^{j(\omega t - \beta_1 z)} \times \frac{E_{i0}}{z_1}\,\check{y}\; e^{j(\omega t - \beta_1 z)}$$

$$\vec{S}_i = \frac{E_{i0}^2}{z_1}\, e^{j(2\omega t - 2\beta_1 z)}\,\check{z} = 2{,}65\ \frac{mW}{m^2}\,\check{z}\; e^{j\left(1{,}26\cdot 10^9\ \frac{1}{s}\, t - 4{,}2 z\right)}$$

→ fasorial:

$$\boxed{\vec{S}_i(z) = 2{,}65\ \frac{mW}{m^2}\,\check{z}\; e^{-j\,4{,}2z}}$$

b) 1° hay que ver cuánto se refleja de los campos (seguro va a ser la mayor parte)

$$\Gamma = \frac{z_2 - z_1}{z_2 + z_1} \approx -0{,}997 + j\,0{,}003 \qquad \rightarrow |\Gamma| \approx 0{,}997$$

$$E_r = |\Gamma|\, E_i = 0{,}997\ \frac{V}{m}$$

$$\vec{E}_r(z,t) = Re\left\{-E_r\,\check{x}\; e^{j(\omega t - \beta_1 z)}\right\}$$

⟶ fasorial:

$$\boxed{\vec{E}_r(z) = -0{,}997\ \frac{V}{m}\,\check{x}\; e^{-j2z}}$$

$$H_r = \frac{E_r}{z_1} = \frac{0{,}997\ \frac{V}{m}}{377\ \Omega} \approx 2{,}64\ \frac{mA}{m}$$

$$\vec{H}_r(z,t) = Re\left\{H_r\,\check{y}\; e^{j(\omega t - \beta_1 z)}\right\}$$

⟶ fasorial:

$$\boxed{\vec{H}_r(z) = 2{,}64\ \frac{mA}{m}\,\check{y}\; e^{-j2z}}$$

$$\vec{S}_r = \vec{E}_r \times \vec{H}_r$$

$$\vec{S}_r = E_r \, \check{x} \, e^{j(\omega t - \beta_1 z)} \times H_r \, \check{y} \, e^{j(\omega t - \beta_1 z)}$$

$$\vec{S}_r = \mathrm{Re}\left\{ E_r H_r \, \check{z} \, e^{2j(\omega t - \beta_1 z)} \right\} = \mathrm{Re}\left\{ 2{,}63 \frac{mW}{m^2} \check{z} \, e^{j\left(1{,}26 \cdot 10^9 \frac{1}{s} t - 4{,}27 z\right)} \right\}$$

$\longrightarrow$ fasorial: $\boxed{\vec{S}_r(z) = 2{,}63 \frac{mW}{m^2} \check{z} \, e^{-j 4{,}27}}$

c) Calculo τ a mano porque sí. Podría usar $\Gamma + 1 = \tau$

$$\tau = \frac{2 z_2}{z_2 + z_1} = 3{,}333\,m + j\,3{,}322\,m \longrightarrow |\tau| = 4{,}706\,m$$

¡medio ②!

$$E_t = E_i |\tau| = 1 \frac{V}{m} \cdot 4{,}706\,m = 4{,}7 \frac{mV}{m}$$

$$\vec{E}_t(z,t) = \mathrm{Re}\left\{ E_t \, \check{x} \, e^{j(\omega t - \beta_2 z)} e^{-\alpha_2 z} \right\} \qquad \alpha_2 = 628{,}3 = \beta_2$$

$\longrightarrow$ f: $\boxed{\vec{E}_t(z) = 4{,}7 \frac{mV}{m} e^{-628{,}3 z} e^{-j 628{,}3 z} \check{x}}$

$$H_t = \frac{E_t}{z_2} = \frac{4{,}7 \frac{mV}{m}}{628{,}3\,m\Omega\,(1+j)} \cong 7{,}49 \frac{mA}{m} \cdot \frac{1}{\sqrt{2} e^{j\frac{\pi}{4}}} = 5{,}30 \frac{mA}{m} e^{-j\frac{\pi}{4}}$$

$$\vec{H}_t(z,t) = \mathrm{Re}\left\{ H_t \, e^{-\alpha z} \, e^{j(\omega t - \beta_2 z)} \, \check{h}_t \right\}$$

$$\vec{H}_t(z,t) = \mathrm{Re}\left\{ 5{,}30 \frac{mA}{m} e^{-j\frac{\pi}{4}} e^{-628{,}3 \frac{z}{m}} \check{y} \, e^{j\left(628{,}3 \cdot 10^6 \frac{1}{s} t - 628{,}3 \frac{1}{m} z\right)} \right\}$$

$\longrightarrow$ fasorial: $\boxed{\vec{H}_t(z) = 5{,}3 \frac{mA}{m} e^{-628{,}3 \frac{1}{m} z} e^{-j 628{,}3 \frac{1}{m} z} \check{y}}$

$$\vec{S}_t = \vec{E}_t \times \vec{H}_t$$

$$\vec{S}_t = E_t\, e^{-\alpha_2 z}\, \check{x}\, e^{j(\omega t - \beta_2 z)} \times H_t\, e^{-\alpha_2 z}\, \check{y}\, e^{j(\omega t - \beta_2 z - \frac{\pi}{4})}$$

$$\vec{S}_t = E_t H_t\, e^{-2\alpha_2 z}\, \check{z}\, e^{2j(\omega t - \beta_2 z)}$$

$\longrightarrow$ f:

$$\boxed{\vec{S}_t(z) = 24{,}94\ \frac{\mu W}{m^2}\, e^{-1257 \frac{z}{m}}\, \check{z}\, e^{-j1257 \frac{z}{m}}\, e^{-j\frac{\pi}{4}}}$$

e)

$$\langle S_i(z=0) \rangle = \frac{1}{2} Re\left\{ 2{,}65 \frac{mW}{m^2}\, \check{z} \right\} = 1{,}326\ \frac{mW}{m^2}\, \check{z}$$

$$\langle S_r(z=0) \rangle = \frac{1}{2} Re\left\{ 2{,}63\ \frac{mW}{m^2}\, \check{z} \right\} = 1{,}317\ \frac{mW}{m^2}\, \check{z}$$

$$\langle S_t(z=0) \rangle = \frac{1}{2} Re\left\{ 24{,}94\ \frac{\mu W}{m^2}\, e^{j\frac{\pi}{4}}\, \check{z} \right\} = \frac{1}{2}\, 25\ \frac{\mu W}{m^2} \cos\left(\frac{\pi}{4}\right) = 8{,}819\ \frac{\mu W}{m^2}\, \check{z}$$

Balance:

$$\langle S_i \rangle - \langle S_r \rangle - \langle S_t \rangle = 1{,}326\ \frac{mW}{m^2} - 1{,}317\ \frac{mW}{m^2} - 8{,}819\ \frac{\mu W}{m^2}$$

$$\approx -5{,}861\ \frac{nW}{m^2} \approx 0$$:)

f) No recuerdo si pedía $\langle P(z) \rangle$ o $\langle P_t(z) \rangle$

Resuelvo asumiendo que pide $\langle P_t(z) \rangle$

$$\langle P_t(z) \rangle = 0, \quad z < 0$$

$z \geq 0$

$$\vec{E}_t \times \vec{H}_t^* = E_t \check{x}\, e^{-\alpha_2 z}\, e^{j(\omega t - \beta_2 z)} \times H_t \check{y}\, e^{-\alpha_2 z}\, e^{-j(\omega t - \beta_2 z - \phi_m)}$$

$$= E_t H_t \check{z}\, e^{-2\alpha_2 z}\, e^{j\phi_m}$$

$$Re\left(\vec{E}_t \times \vec{H}_t^*\right) = 24{,}94 \frac{\mu W}{m^2}\, e^{-1257 \frac{z}{m}} \cos\left(\frac{\pi}{4}\right) \check{z}$$

$$\langle P_t(z) \rangle = \frac{1}{2} Re\left(\vec{E}_t \times \vec{H}_t^*\right) = \frac{\sqrt{2}}{2} \cdot 12{,}47 \frac{\mu W}{m^2}\, e^{-1257 \frac{1}{m} z}\, \check{z}$$

$$\langle P_t(z) \rangle = 8{,}819 \frac{\mu W}{m^2}\, e^{-1257 \frac{1}{m} z}\, \check{z}$$

$$\Rightarrow \boxed{\langle P_t(z) \rangle = 8{,}819 \frac{\mu W}{m^2}\, e^{-1257 \frac{1}{m} z}\, \check{z}\, \mathbb{1}\{z \geq 0\}}$$

$$1\,dB \equiv 10 \log\left(\frac{P}{P_{ref}}\right)$$

$$10 \log\left(\frac{\langle P_t(z) \rangle}{\langle P_t(z=0) \rangle}\right) < -100$$

$$\Leftrightarrow \log\left(\frac{\langle P_t(z) \rangle}{\langle P_t(0) \rangle}\right) < -10 \quad \Leftrightarrow \quad \frac{\langle P_t(z) \rangle}{\langle P_t(0) \rangle} < 10^{-10}$$

$$\Leftrightarrow \langle P_t(z) \rangle < 10^{-10} \langle P_t(0) \rangle$$

Reemplazo para despejar:

$$8{,}819\ \frac{\mu W}{m^2}\, e^{-1257\ \frac{1}{m} z}\ \check{z}\ \mathbb{1}\{z \geq 0\} < 10^{-10}\ \ 8{,}819\ \frac{\mu W}{m^2}\ \check{z}$$

$z \geq 0$

$$\Leftrightarrow\quad e^{-1257\ \frac{1}{m} z} < 10^{-10} \qquad \overset{\ln(\cdot)\ \text{m.a.m}}{\Leftrightarrow} \quad -1257\ \frac{1}{m}\, z < \ln\left(10^{-10}\right)$$

$$\Leftrightarrow \quad z > \frac{\ln(10^{-10})}{-1257\ \frac{1}{m}} \quad \therefore \quad \boxed{z > 18{,}32\ mm}$$

Chequeo:

$$10 \log\left(\frac{8{,}819\ \frac{\mu W}{m^2}\, e^{-1257\ \frac{1}{m} \cdot 18{,}32\ mm}}{8{,}819\ \frac{\mu W}{m^2}}\right) = 10 \log\left(e^{-23{,}03}\right) = -100\ dB$$

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